

# **Milestone-3 Inline Inference and Rejection Pipeline**

The Python script automates **pill image defect detection using a Roboflow model** and stores the prediction results in a CSV file while avoiding re-processing previously analyzed images.

## **Workflow Overview**

### **1. Configuration Setup**

- Defines:
  - Image folder location
  - Roboflow model ID (new-pill/2)
  - API key
  - Output CSV file path (prediction\_results.csv)

### **2. Check Previously Processed Images**

- Reads the existing CSV file (if available).
- Extracts all processed image names.
- Creates a list of images that have **not yet been analyzed**.
- Prevents duplicate predictions and saves processing time.

### **3. Scan Image Directory**

- Searches for all .jpeg, .jpg, and .png files.
- Filters out images already present in the CSV log.
- Sorts the remaining new images.

### **4. Initialize Roboflow Inference Client**

- Connects to the Roboflow hosted detection API using the provided API key.

### **5. Process Images One by One**

For each new image:

- Generates:
  - Unique Image ID (IMG\_0001, IMG\_0002, etc.)
  - Current timestamp
  - Batch number (5 images per batch)
- Sends the image to the Roboflow model for inference.
- Retrieves defect predictions.

### **6. Handle No-Detection Cases**

- If no defect is detected:
  - Creates a default record with:
    - Class = "None"
    - Confidence = 0.0
    - Bounding box coordinates = 0

## 7. Create Structured Dataset

- Converts prediction results into a Pandas DataFrame.
- Adds metadata:
  - Batch number
  - Image ID
  - Image name
  - Timestamp
- Keeps only required columns.

## 8. Append Results to CSV

- Writes prediction results to prediction\_results.csv.
- Uses append mode so previous records remain intact.
- Adds headers only when creating the file for the first time.

## 9. Error Handling

- If inference fails for any image, the exception is caught and displayed.
- Processing continues for the remaining images.

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### Output CSV Format

| batch_number | image_id | image_name | timestamp           | class   | confidence | x   | y  | width | height |
|--------------|----------|------------|---------------------|---------|------------|-----|----|-------|--------|
| 1            | IMG_0001 | pill1.jpg  | 2026-06-10 10:30:15 | Scratch | 0.95       | 120 | 80 | 40    | 35     |
| 1            | IMG_0002 | pill2.jpg  | 2026-06-10 10:30:20 | None    | 0.00       | 0   | 0  | 0     | 0      |

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### Sample CSV File

| batch_number | image_id   | image_name                    | timestamp        | class                  | confidence | x     | y     | width | height |
|--------------|------------|-------------------------------|------------------|------------------------|------------|-------|-------|-------|--------|
|              | 1 IMG_0001 | ChatGPT Image Jun 9, 2026, 03 | 10-06-2026 10:04 | faulty_print           | 0.925376   | 638.5 | 683.5 | 567   | 379    |
|              | 1 IMG_0002 | ChatGPT Image Jun 9, 2026, 03 | 10-06-2026 10:05 | faulty_print           | 0.760712   | 630.5 | 668   | 529   | 342    |
|              | 1 IMG_0001 | ChatGPT Image Jun 9, 2026, 10 | 10-06-2026 10:12 | coating_irregularities | 0.913574   | 918   | 874.5 | 184   | 157    |
|              | 1 IMG_0001 | ChatGPT Image Jun 9, 2026, 10 | 10-06-2026 10:12 | coating_irregularities | 0.888159   | 542.5 | 634   | 183   | 122    |
|              | 1 IMG_0002 | ChatGPT Image Jun 9, 2026, 10 | 10-06-2026 10:12 | coating_irregularities | 0.908053   | 487.5 | 389   | 193   | 154    |
|              | 1 IMG_0002 | ChatGPT Image Jun 9, 2026, 10 | 10-06-2026 10:12 | coating_irregularities | 0.869304   | 996   | 865   | 114   | 122    |
|              | 1 IMG_0003 | ChatGPT Image Jun 9, 2026, 10 | 10-06-2026 10:13 | unknown_tablet         | 0.769565   | 625   | 601   | 440   | 452    |
|              | 1 IMG_0004 | ChatGPT Image Jun 9, 2026, 10 | 10-06-2026 10:13 | unknown_tablet         | 0.56936    | 629   | 616.5 | 496   | 247    |
|              | 1 IMG_0004 | ChatGPT Image Jun 9, 2026, 10 | 10-06-2026 10:13 | unknown_tablet         | 0.554622   | 640   | 607.5 | 642   | 439    |
|              | 1 IMG_0005 | ChatGPT Image Jun 9, 2026, 10 | 10-06-2026 10:13 | unknown_tablet         | 0.570705   | 628.5 | 612.5 | 481   | 431    |
|              | 2 IMG_0006 | ChatGPT Image Jun 9, 2026, 10 | 10-06-2026 10:13 | unknown_tablet         | 0.645991   | 634   | 609.5 | 566   | 329    |
|              | 2 IMG_0006 | ChatGPT Image Jun 9, 2026, 10 | 10-06-2026 10:13 | unknown_tablet         | 0.52841    | 639   | 567.5 | 626   | 431    |
|              | 2 IMG_0007 | ChatGPT Image Jun 9, 2026, 11 | 10-06-2026 10:13 | scratch                | 0.869501   | 409   | 395.5 | 264   | 147    |
|              | 2 IMG_0008 | ChatGPT Image Jun 9, 2026, 11 | 10-06-2026 10:14 | scratch                | 0.842543   | 910.5 | 447.5 | 117   | 283    |